



Princeton Computer Science Contest – Fall 2025

Problem 7: PSET Groups (18 points) [File Upload]

By Andrew Tu

In COS 675, students are allowed to work together on the problem sets, in groups of up to 3 students. As the course admin, you are tasked with assisting the formation of the groups. In particular, you survey the students in the class, who each respond with a list of other students they would like to work together with.

- If 2 students each want to work together with the other, this pairing is considered **cooperative**.
- If 3 students are all pairwise cooperative, this is considered a **good** group.

Part 1 (2 points)

The survey has been out for 1 day, and so far 8 people have responded. From their responses, you find exactly 18 cooperative pairs. What is the minimum number of good groups that could be formed and how could this be achieved?

Submit only the integer answer (in a text file); no proof is required; at most 2 attempted submissions.

Part 2 (6 points)

It's been a couple of days. Last you checked, n students have responded and among their responses there are exactly m cooperative pairs. Unfortunately, given your forgetful nature, you forgot which pairs of students are cooperative. Hence, for now, you must assume that each pair of students is equally likely to be cooperative.

Let G be the expected number of good groups that could be formed from the survey results, and let $\rho = \frac{m}{n}$. For a fixed ρ , as n and m tend to infinity, G tends $\rho^3 \cdot \alpha$ for some constant α . Show that G indeed tends to $\rho^3 \cdot \alpha$ and find α .

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Part 3 (10 points)

Disaster has struck: eduroam is down! You are unable to see the survey results, nevertheless, the University wants you to assess how cooperative the class is as a whole by seeing how many groups could be formed. In particular, you need to find a lower bound on the number of groups that could be formed.

When eduroam comes back up, you don't have time to iterate through every potential good group. Rather, you come up with the following lower bound:

For n respondents with exactly m cooperative pairs, show that there are at least $f(n, m)$ good groups, where

$$f(n, m) = \frac{m}{3n}(4m - n^2)$$

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